

University of Illinois at Chicago
DEPARTMENT OF CHEMICAL ENGINEERING

CHE 433 Homework Set #5

Distillation Column
Integration in the
Production of Cyclohexane

Prepared for:
Process Simulation with Aspen Plus
Dr. Michael Caracotsios
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Prepared by:
Courteney Fabbri
Kellen Krupa

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TABLE OF CONTENTS

TABLE OF CONTENTS.....	2
EXECUTIVE SUMMARY.....	3
SYSTEMS AND PROCEDURES.....	4
RESULTS.....	7
CONCLUSION.....	7
APPENDICES.....	8
Appendix I: Equations/Formulas.....	8
Appendix II: PFD.....	10
Appendix III: Block Report.....	11
REFERENCES.....	14

EXECUTIVE SUMMARY:

The hydrogenation of benzene was simulated in ASPEN to maximize the recovery of cyclohexane for the production of 315 KMTA of cyclohexane. Unlike the production process designed in Assignment #3, the flowsheet was reconfigured using unit operations that are closer to actual equipment in the field. Using Assignment #4 as the base simulation, a shortcut simulation was performed to find the operating reflux ratio, number of stages, feed stage location, and the distillate to feed ratio. This information was then used in a rigorous simulation that obtained a 3 ft column diameter and a 99.9967% recovery of cyclohexane. Lastly, operating costs and utilities were calculated using economic analysis formulas for each unit.

SYSTEMS AND PROCEDURES:

Solving for the maximum recovery of cyclohexane required the use of shortcut and rigorous simulations in order to find operating data and distillation column information. Beginning with the simulation from Assignment #4, a shortcut model was performed to find the operating reflux ratio, number of stages, feed stage location, and the distillate to feed ratio. Table 1 presents the specified reflux ratio, light key (LK), heavy key (HK), and condenser and reboiler pressures. Cyclohexane was selected as the heavy key because the objective is to obtain a maximum recovery of this component. A partial vapor condenser was selected under “condenser specifications”. After running the simulation, the previously mentioned data was obtained and input into the rigorous model.

Table 1. Specified inputs for shortcut simulation.

Reflux Ratio	-1.15
LK	Methane
HK	Cyclohexane
Condenser Pressure (psia)	14
Reboiler Pressure (psia)	15

To determine the diameter of the column, a tray sizing specification was added and the simulation was run again. The results were available in the block’s results section. Next, the efficiency of the column was calculated using the O’Connell equation (equation 2 Appendix I). Obtaining the K-values and viscosity of the feed stream, a property set was created and the simulation was run. Results were available in the stream results table. Using the K-values, α was calculated. Then the efficiency and number of trays were calculated.

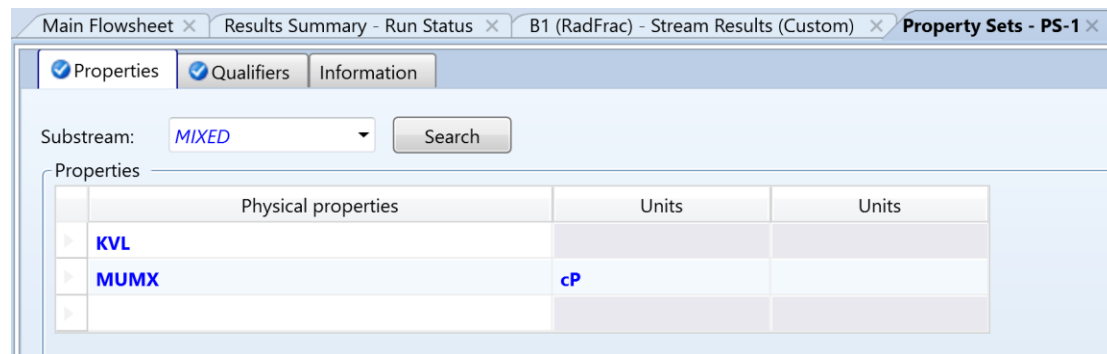


Figure 1. Property set specification.

Qualifiers of selected properties			
Phase	Liquid	Vapor	
Component	CH	C1	
2nd liquid key component			
Temperature	☒ System	F	
Pressure	☒ System	psia	
% Distilled			
Water basis			
Base component			
Component group			
Base component group			

Boiling point range: to F

Base boiling point range: to F

Figure 2. Property set specification.

Cost analysis of the materials used in the rigorous model RADFRAC were completed using several different economic formulas for different block variables. The distillation column—comprised of the vertical column, 19 sieve trays, a partial condenser, a kettle reboiler, and a reflux pump—was economically analyzed to calculate cost of materials, installed cost, and annual cost.

To begin, the O’Connell (Equation 2) was used to determine the number of trays in the column. Using a calculated efficiency of 49%, the number of trays was determined to be 19. From this, the height of the tower (tangent to tangent) was determined to be approximately 9.75 meters. Using the diameter determined in RADFRAC, the mass of the shell of the vessel was determined and used with Equation 9 to calculate the cost of the vertical vessel.

Next, the diameter (0.9455 m) was used to calculate the price of the sieve trays. Using Equation 11, the cost of all 19 sieve trays was determined. After that, the condenser material cost was determined using Equation 3. The heat exchanger’s area (2.0058 m²) was used in this calculation. The next component to be analyzed was the kettle reboiler, which was determined using Equation 12 and an area of 11.53 m². Finally, the reflux pump was calculating using a flow rate of 10.1991 L/s in conjunction with Equation 8.

A	B	C	D	E	F	G
Cyclohexane Production Cost Estimate						
EQUIPMENT LIST: CAPITAL COSTS						
Equipment Class	Project: Assignment #2	Chemical Engineering Department	Service Description	Tag	Material Cost	Installed Cost
	Location:					
	ASPEN BlockID					
	Distillation					
	Vertical column	xxxx	Distillation column		\$44,476	\$177,905
	Trays	xxxx	Trays in column (x19)		\$14,039	\$35,097
Condenser	xxxx		Stage 1 heat exchanger		\$39,374	\$137,810
Reboiler	xxxx		Kettle reboiler		\$45,656	\$91,313
Reflux pump	xxxx		Reflux pump		\$13,917	\$55,667
						\$74,223
						\$663,722

Figure 3. Capital cost of equipment

A	B	C	D	E	F	G	H
Cyclohexane Production Cost Estimate							
UTILITIES: OPERATING COST							
Utility Class	Project: Assignment #2	Chemical Engineering Department	Service	Usage	Usage Units	Unit Cost	Annual Cost, \$/yr
	Location:						
	ASPEN ID						
	Electricity		Reflux pump	11.19 kWh		\$0.09	\$8,822 kW/year
	Cooling water		Coolant in condenser	157.20 US gal/hour		\$0.08	\$110,166 US gal/year
	High pressure steam		Heats reboiler	2.76 kb/hour		\$11.45	\$276,523 kb/year
Fuel gas			Byproduct	74,277.57 MMBtu		(\$7.30)	-\$542,226 MMBtu/yr
Benzene			Raw material feed	325.02 Mtons/year		\$760	\$247,012,540 tons/year
Hydrogen			Raw material feed	28,531.48 MSCFD		\$2.60	\$27,076,375 MSCFY
Cyclohexane			Product	347.08 Mtons/year		(\$1,040)	-\$360,959,144 tons/year
							-\$87,016,945 PROFIT

Figure 4. Utility costs of production

RESULTS:

Table 2. Data from shortcut (DISTWU) simulation.

		Unit
Minimum reflux ratio:	0.48	
Actual reflux ratio:	0.55	
Minimum number of stages:	2.21	
Number of actual stages:	8.17	
Feed stage:	4.85	
Number of actual stages above feed:	3.85	
Reboiler heating required:	2036761.19	Btu/hr
Condenser cooling required:	139587.47	Btu/hr
Distillate temperature:	-78.47	F
Bottom temperature:	180.42	F
Distillate to feed fraction:	0.02	

Table 3. Rigorous (RADFRAC) Distillation.

Column diameter	3 ft
α (relative volatility)	2
μ (viscosity)	0.52
η (efficiency)	49.87%
# Trays	19

Table 4. Cost Summary.

Distillation	Material Cost	Installed Cost	Annual Cost, \$/yr
Vertical column	\$44,476	\$177,905	\$237,206
Trays	\$14,039	\$35,097	\$46,797
Condenser	\$39,374	\$137,810	\$183,747
Reboiler	\$45,656	\$91,313	\$121,750
Reflux pump	\$13,917	\$55,667	\$74,223

CONCLUSION:

A target cyclohexane production rate of 312 KMTA was accomplished. Integration of real unit operations into the flowsheet was successful in saving operating costs, specifically for the heat integration of heat exchangers HEAT1 and HEAT2. Cooling water provided to be a cheap and efficient way to cool the effluent of the reactor. Total capital cost for the production of 312 KMTA of cyclohexane is \$956,221 per year, while total utilities (including fuel credit) were found to be \$740,078,652 given the huge expense of raw material cost of benzene and hydrogen at \$740,098,410 per year. Therefore, costs could be greatly reduced from raw materials alone as cooling water did not provide much of an expense. Furthermore, a cheaper compressor could save hundreds of thousands of dollars as it was by far the most expensive unit block.

APPENDIX I: EQUATIONS/FORMULAS

Recovery:

(Equation 1)

$$recovery = \frac{composition\ in\ PRODUCT}{composition\ in\ FEED}$$

Efficiency of a column O'Connell equation:

(Equation 2)

$$\eta = 50.3(\alpha * \mu_L)^{-0.226}$$

Number of trays:

(Equation 2.5)

$$\# trays = \frac{\# stages}{\eta/100}$$

Cost to purchase a heat exchanger:

(Equation 3)

$$\$ (2020) = [28000 + 54[A(m^2)]^{1.2}] * 1.4$$

Marshall and Swift (M&S) index:

(Equation 4)

$$\$ (2020) = \$ (2010) * \frac{index(2020)}{index(2010)}$$

Area of a heat exchanger:

(Equation 5)

$$Q = F_T A U \Delta T_{ln}$$

Logarithmic mean temperature difference:

(Equation 6)

$$\Delta T_{ln} = \frac{\Delta T_1 - \Delta T_2}{\ln \left(\frac{\Delta T_1}{\Delta T_2} \right)}$$

Annualized cost:

(Equation 7)

$$\text{Annualized} = \left(\frac{4}{3}\right) * (\text{Installed cost})$$

Cost to purchase a pump:

(Equation 8)

$$\$ (2020) = [8000 + 240\{Q(L/s)\}^{0.9}] * 1.4$$

Cost to purchase a vertical vessel:

(Equation 9)

$$\$ (2020) = [11600 + 34\{M(kg)\}^{0.85}] * 1.4$$

Mass of vertical vessel:

(Equation 10)

$$M = (\pi DL * \text{thickness}) \rho_{\text{steel}}$$

Cost to purchase sieve trays:

(Equation 11)

$$\$ (2020) = [130 + 440[\text{Diameter}(m)]^{1.8}] * 1.4$$

Cost to purchase a kettle reboiler:

(Equation 12)

$$\$ (2020) = [29000 + 400(\text{Area}(m^2))^{0.9}] * 1.4$$

Figure i. Shortcut (DISTWU) Simulation

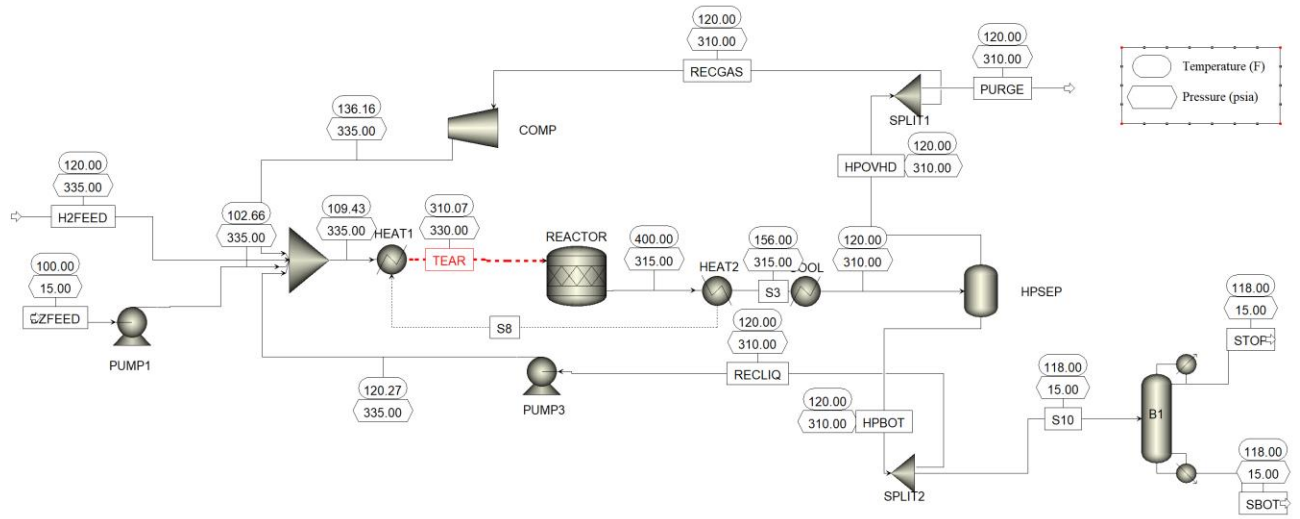
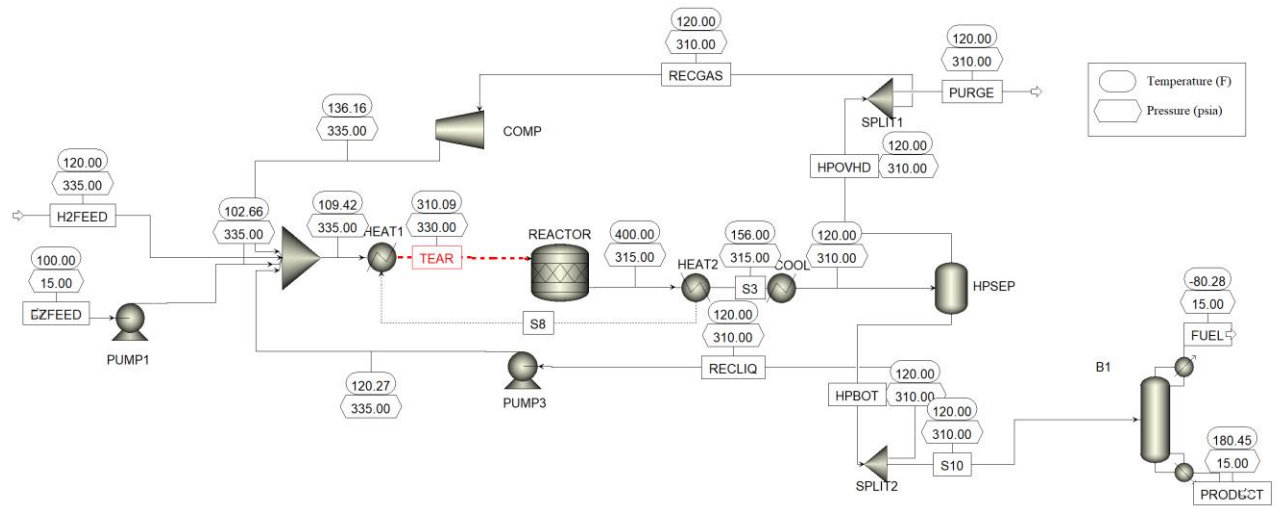


Figure ii. Rigorous (RADFRAC) Simulation



APPENDIX III: STREAM RESULTS

Shortcut (DISTWU) Simulation

	S10	STOP	SBOT
Substream: MIXED			
Mole Flow lbmol/hr			
H2	5.62	5.62	0.00
N2	1.23	1.23	0.00
C1	14.80	14.80	0.00
BZ	1.28	0.00	1.28
CH	940.54	0.01	940.53
WATER	0.00	0.00	0.00
Mole Frac			
H2	0.01	0.26	0.00
N2	0.00	0.06	0.00
C1	0.02	0.68	0.00
BZ	0.00	0.00	0.00
CH	0.98	0.00	1.00
WATER	0.00	0.00	0.00
Total Flow lbmol/hr	963.47	21.65	941.82
Total Flow lb/hr	79540.71	283.90	79256.81
Total Flow cuft/hr	1715.21	6310.13	1770.43
Temperature F	120.00	-78.47	180.42
Pressure psia	310.00	14.00	15.00
Vapor Frac	0.00	1.00	0.00
Liquid Frac	1.00	0.00	1.00
Solid Frac	0.00	0.00	0.00
Enthalpy Btu/lbmol	-64413.15	-23236.47	-63048.92
Enthalpy Btu/lb	-780.23	-1772.05	-749.22
Enthalpy Btu/hr	-62060000.00	-503080.00	-59381000.00
Entropy Btu/lbmol-R	-140.76	-14.25	-139.64
Entropy Btu/lb-R	-1.71	-1.09	-1.66
Density lbmol/cuft	0.56	0.00	0.53
Density lb/cuft	46.37	0.04	44.77
Average MW	82.56	13.11	84.15
Liq Vol 60F cuft/hr	1642.81	18.58	1624.22

Rigorous (RADFRAC) Simulation

	Units	FUEL	PRODUCT	S10
From		B1	B1	SPLIT2
To				B1
Substream: MIXED				
Phase:		Vapor	Liquid	Liquid
Component Mole Flow				
H2	KMOL/HR	2.547	0.000	2.55
N2	KMOL/HR	0.557	0.000	0.56
C1	KMOL/HR	6.712	0.000	6.71
BZ	KMOL/HR	0.000	0.582	0.58
CH	KMOL/HR	0.004	426.581	426.58
WATER	KMOL/HR	0.000	0.000	0.00
Component Mole Fraction				
H2		0.259	0.000	0.01
N2		0.057	0.000	0.00
C1		0.684	0.000	0.02
BZ		0.000	0.001	0.00
CH		0.000	0.999	0.98
WATER		0.000	0.000	0.00
Mole Flow	KMOL/HR	9.820	427.163	436.98
Mass Flow	KTONNE/YEAR	1.128	315.112	316.24
Volume Flow	L/MIN	2765.473	835.496	809.42
Temperature	K	210.774	355.624	322.04
Pressure	ATM	1.021	1.021	21.09
Vapor Fraction		1.000	0.000	0.00
Liquid Fraction		0.000	1.000	1.00
Solid Fraction		0.000	0.000	0.00
Molar Enthalpy	CAL/MOL	-12916.210	-35026.450	-35785.09
Mass Enthalpy	CAL/GM	-985.349	-416.223	-433.46
Enthalpy Flow	CAL/SEC	-35231.080	-4156100.000	-4343700.00
Molar Entropy	CAL/MOL-K	-14.417	-139.643	-140.76
Mass Entropy	CAL/GM-K	-1.100	-1.659	-1.71
Molar Density	MOL/CC	0.000	0.009	0.01
Mass Density	GM/CC	0.001	0.717	0.74
Average Molecular Weight		13.108	84.153	82.56
Phase: All				
K-values, vapor/liquid				
CH		2.150	0.964	0.02
C1		60.038	336.682	7.08
Phase: Vapor				

Viscosity, mixture	CP	0.009		
Phase: Liquid				
Viscosity, mixture	CP		0.4071533	0.52

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